# Aboabo, Kumasi

*Every death on Aboabo's record is a drowning in the drains at the first rains, which come straight after the February–March heat peak. Ashanti's forecast heat deaths have never been counted here. Start a named heat list and a daily death count by February, and clear the Pelele and Asabi drains by 15 February.*

| REGION | EXCESS HEAT DEATHS | WINDOW | DOMINANT HAZARD |
|---|---|---|---|
| Ashanti Region (GADM), region level | 1,000 forecast Oct 2026 – Mar 2027, 700 of them in Feb–Mar; 71 already past (Jun–Sep) | 6-month forecast, Oct 2026 – Mar 2027 | Humid heat Feb–Mar in shared rooms; drownings at the first rains; power and water cuts; cholera |

**28/28 recommendations checked.** 11 confirmed, 16 corrected, 0 unverified, 1 contraindicated (9 actions, all corrected; 10 "Do not" cards: 6 confirmed, 3 corrected, 1 contraindicated; 9 For you steps: 5 confirmed, 4 corrected). Exposure figure: Climate Impact Lab El Niño Heat Mortality Tracker data, 26 October 2026, pre-peer-review, for Ashanti Region, heat only; nothing published covers Aboabo or the municipality itself. The 1,100 is model-driven: at Kintampo, the surveillance site closest to Kumasi's climate, death risk rises at both extremes, in the cold as well as the heat. The funding target did not move in verification: GHS 770,000 (US$66,094).

EXECUTIVE SUMMARY — AT A GLANCE




6.70°N 1.60°W, Ghana · Natural Earth

- **The risk.** Every recorded death in Aboabo is a drowning in the drains at the first rains; the forecast heat deaths have never been counted.
- **For you.** From the first heavy rain in March, never cross moving water, and keep children and porters off the Pelele and Asabi drains.
- **The money.** Floor GHS 270,000, target 770,000, ceiling 1.445M (US$23,176 / 66,094 / 124,034); Red Cross heat volunteers buy the most.
- **Do not.** Do not copy the "no fans above 35 °C" rule: in Kumasi's humid heat a fan plus wet skin cools older people.*


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* Suggested formulation: check with a health worker before sharing.

FOR YOU, AS A RESIDENT 9 STEPS

For a household in Aboabo, this winter brings the hottest, most humid weeks in shared compound rooms in February and March, power cuts that stop the water pumps, Ramadan from about 8 February, and then the first March storms, which are what have killed people here. The ranked actions below are written for the Assembly and its partners. These are the ones you can take yourself.

### Keep three days' water in lidded containers, filled in December–January at the lifeline rate

PARTIALLY CONFIRMED FROM ACTION 4 AND "DO NOT" 7

| BY WHEN | COST TO YOU | PROTECTS |
|---|---|---|
| filled during December 2026 – January 2027 | water at GHS 5.99 (US$0.51) a cubic metre for the first 5 m³; no container price was verified | drinking water through the pump cuts, and the household from cholera and mosquitoes |

Since 1 July 2026 the lifeline rate for the first 5 m³ has been GHS 5.99 (US$0.51) a cubic metre, and PURC held it unchanged for the fourth quarter. In February 2026 power cuts slashed output at the Barekese and Owabi plants, which supply most of Kumasi, and parts of the city went three days without water. Cover every container: open water breeds *Aedes* mosquitoes and carries a cholera risk.

### Keep ORS sachets at home; at any diarrhoea, start ORS and go to the clinic the same day

CONFIRMED FROM ACTION 5

| BY WHEN | COST TO YOU | PROTECTS |
|---|---|---|
| stocked by 31 January 2027 | a few sachets (no Kumasi retail price was verified) | young children and older people |

ORS cuts diarrhoea deaths in children under five by about 69% (Munos and colleagues, 2010), and zinc by about 23% (Fischer Walker and Black, 2010). Dry taps push people onto stored and bought water, which is when cholera spreads; Ashanti had 203 cases in 2014.*

### On hot nights, run the fan and wet the skin; when the power is off, use a damp cloth and an open window

CONFIRMED FROM "DO NOT" 4 AND 8

| BY WHEN | COST TO YOU | PROTECTS |
|---|---|---|

* Suggested formulation: check with a health worker before sharing.

| | | |
|---|---|---|
| February–March 2027 | GHS 0 with a fan you already have | older people, and anyone with heart disease |

At 38 °C and 60% humidity, fans plus skin wetting lowered heart strain in older adults, including people with heart disease (*NEJM* 2024); fans did harm only in hot, dry air. Kumasi's February maximum is about 33–34 °C, under every fan threshold. Do not add ceiling insulation for night relief: in Accra it made rooms 1.4 °C warmer at night.*

### From the first heavy rain in March, never cross moving water; pick your crossing in daylight; keep children and porters off the Pelele and Asabi drains in rain and after dark

CONFIRMED · FROM ACTION 6 AND "DO NOT" 10

| BY WHEN | COST TO YOU | PROTECTS |
|---|---|---|
| agree the route by 1 March 2027; keep to it through June | GHS 0 | children, porters, and anyone walking home in a storm |

On 10 March 2008, in the season's first heavy rain, two teenage head porters fell into the same Aboabo drain, the second trying to save the first; one died in hospital and the other's body was never found. In June 2021 people drowned at Aboabo and Asabi Junction. The reliable rains in the forest zone start in late March, but the 2008 deaths came on 10 March.

### Ramadan, about 8 February – 9 March 2027: the sick, elderly, and pregnant are exempt; drink between iftar and suhoor; a wet cloth on the neck

CONFIRMED · FROM ACTION 3

| BY WHEN | COST TO YOU | PROTECTS |
|---|---|---|
| from about 8 February 2027, depending on the moon | GHS 0 | older and sick people, pregnant women, and outdoor workers |

Qur'an 2:184 exempts the sick, the elderly, and pregnant women from the fast. The fast falls on the two worst forecast months for Ashanti (360 deaths in February, 340 in March). In February 2026 the only heat advice for fasting Muslims in Ghana came from a Tema NGO, and none was aimed at Kumasi.*

* Suggested formulation: check with a health worker before sharing.

### If someone is due February–April, tell the antenatal clinic, and keep her cool and rested in the last weeks

PARTIALLY CONFIRMED · FROM ACTION 1

| BY WHEN | COST TO YOU | PROTECTS |
|---|---|---|
| tell the clinic by 31 January 2027 | maternal care is free under the NHIS, but informal charges are still reported | the mother and the baby at birth |

In 138,015 births at 16 hospitals in Benin, Malawi, Tanzania, and Uganda, heat in the week before birth raised the risk of perinatal death (odds ratio 1.34, 95% CI 1.01–1.78; Hanson and colleagues, *Nature Medicine* 2024); stillbirth alone was not significant. NHIS free maternal care has been in place since 2008, but emergency obstetric care is not fully covered, so ask the clinic what you will pay.*

### Name one adult in each compound to check on the elderly, babies, and the sick on hot nights and in power cuts

PARTIALLY CONFIRMED · FROM ACTION 1

| BY WHEN | COST TO YOU | PROTECTS |
|---|---|---|
| agreed by 31 January 2027; checks February–March | GHS 0 | people who would die unseen in a closed room |

About 37% of adults screened in the Ashanti Region had high blood pressure, and about 1 in 8 Kumasi newborns carried the sickle-cell trait (screening, 1995–2004), and heat bears hardest on people with these conditions. No list of who is at risk exists in Aboabo, so the compound is the unit that can see them.

### Ask the Social Welfare office whether your household qualifies for LEAP, by 15 December

PARTIALLY CONFIRMED · FROM ACTION 8

| BY WHEN | COST TO YOU | PROTECTS |
|---|---|---|
| 15 December 2026 | free to enrol | household cash through the hot season |

From September 2026 LEAP pays GHS 380–630 (US$33–54) per household every two months, depending on household size. No precedent was found for paying a cycle early in a shock, so enrolment, not a top-up, is what you can count on.

* Suggested formulation: check with a health worker before sharing.

### Know where alerts come from: 112 for emergencies, GMet forecasts, and Haske FM 91.7 in Hausa

CONFIRMED FROM ACTIONS 1 AND 3

| BY WHEN | COST TO YOU | PROTECTS |
| --- | --- | --- |
| now | GHS 0; 112 is toll-free | everyone in the compound |

112 is the national toll-free emergency number, open 24 hours, and NADMO promotes it for floods. GMet publishes forecasts and warnings at meteo.gov.gh; it issues seasonal heat advisories but no heat-index threshold. Haske FM 91.7 is a Kumasi community station that broadcasts mainly in Hausa, with some Twi and English.*

## What to ask of the people in charge

- Ask the Municipal Health Directorate: "Who is on our area's heat list, and who visits them?" (action 1).
- Ask ECG and Ghana Water: "Will the Barekese and Owabi pumps get priority feeders, and when will the rationing schedule be published?" (action 4).
- Ask the MCE: "Will the Pelele and Asabi drains be cleared, and the silt hauled away, by 15 February?" (action 6).
- Ask NAFCO and MoFA: "What price rise releases buffer stock into Kumasi markets?" (action 8).

WHAT THIS PLACE SHOULD DO RANKED

Ranked by expected lives saved per unit of effort before the peak. Backing lenses are shown per action; an action several lenses reached

* Suggested formulation: check with a health worker before sharing.

independently is the stronger bet.

## 1. A named list and a heat check, February–March

PARTIALLY CONFIRMED | PUBLIC HEALTH · EMERGENCY MANAGER · FRONTLINE · PHILANTHROPIST | ADAPTATION INVENTORY: NO PRECEDENT CONFIRMED

| OWNER | BY WHEN | COST | PREVENTS | PAYER / TRIGGER |
|---|---|---|---|---|
| Asokore Mampong Municipal Health Directorate, with community health volunteers, the Ghana Red Cross Society (Ashanti), and mosque networks | list by 31 January 2027; visits and shaded yards February–March | GHS 75,000 (US$6,438) for 40 Red Cross volunteers over 11 weeks (funding placement 1) | older and sick people dying unseen in hot shared rooms at the forecast peak | donor grant to the Red Cross; trigger set locally, for example on GMet's daily forecast for Kumasi |

By 31 January, list the elderly, people with hypertension, diabetes, or sickle-cell disease, the bedridden, and infants living in compound rooms. On hot days in February and March, visit them, and open shaded mosque and school yards from 1 to 4 pm. Ashanti is forecast 360 heat deaths in February and 340 in March. The model is Ahmedabad's Heat Action Plan, which averted about 1,190 deaths a year (95% CI 162–2,218; Hess and colleagues, *Journal of Environmental and Public Health* 2018). In an Ashanti community screening 37.4% of adults had hypertension (Tannor and colleagues, an opportunistic sample), and Kumasi's newborn screening found the sickle-cell trait in 13.3% of babies (1995–2004).

* Suggested formulation: check with a health worker before sharing.

Corrected in verification: GMet publishes no heat-index threshold, and the Ghana Health Service has no heat warning system, only general advice; its chief forecaster has said a heatwave is not a recognised phenomenon in Ghana. The trigger must therefore be set locally, and that gap should be named to the Directorate. Ahmedabad's heat is dry and peaks above 45 °C, so it transfers to humid Kumasi (about 33–34 °C) as a way to organise a response, not as an effect size. The hypertension figure is for the Ashanti Region, not Kumasi alone.

## 2. Count deaths every day from 1 February: burials and heat admissions

PARTIALLY CONFIRMED | PUBLIC HEALTH · HISTORIAN | ADAPTATION INVENTORY: NO PRECEDENT CONFIRMED

| OWNER | BY WHEN | COST | PREVENTS | PAYER / TRIGGER |
|---|---|---|---|---|
| Municipal Health Directorate, with imams and zongo cemetery keepers (daily burial count), and Manhyia District Hospital and Komfo Anokye Teaching Hospital (heat-illness admissions) | daily from 1 February 2027 | about GHS 0: tally sheets and existing staff | a peak that passes uncounted, and actions nobody can judge | no payer needed; trigger: the date |

Kumasi has never counted a heat death, and Ghana's death registration is incomplete, so nobody would know whether the forecast deaths happened. Muslim burials in the zongo take place within about 24 hours, so a daily count kept by imams and cemetery keepers gives a near-real-time signal. Add the Tafo cemetery or the municipal register, because the imams' count misses Christian burials. Burial counts have been used as mortality surveillance in Mogadishu (satellite images checked against cemetery registers), Addis Ababa (gravedigger reports), and Kano (gravedigger alerts).

* Suggested formulation: check with a health worker before sharing.

Corrected in verification: feasible, with African precedent, but no Ghana or Kumasi precedent was found.

## 3. A Friday sermon on fasting in the heat, 5 February 2027

PARTIALLY CONFIRMED | FRONTLINE · PUBLIC HEALTH | ADAPTATION INVENTORY: NO PRECEDENT CONFIRMED

| OWNER | BY WHEN | COST | PREVENTS | PAYER / TRIGGER |
|---|---|---|---|---|
| Office of the Ashanti Regional Chief Imam, Sheikh Abdul Rahman Zakaria Al Shoukry, with the zongo chief imams and Haske FM 91.7 for a call-in | Friday 5 February 2027, the Friday before Ramadan is expected | volunteer time | fasting older, sick, and pregnant people dehydrating at the heat peak | no payer needed; trigger: the date |

Give the imams one page in Hausa and Twi: the Qur'anic exemption for the sick, elderly, and pregnant (2:184), drinking between iftar and suhoor, a wet cloth on the neck, and the danger signs of heat illness. Run the same script as a call-in on Haske FM 91.7, a Kumasi community station broadcasting mainly in Hausa. Ramadan is expected from about 8 February (the evening of 7 February) to 9 or 10 March 2027, depending on the moon, so it lands on Ashanti's two worst forecast months. The channel exists: in October 2022 about 125 Ashanti zongo chief imams formed the ZOLAOD coalition on sanitation. In February 2026 the only heat advice to fasting Muslims came from Eco Impact Network, an NGO in Tema.*

* Suggested formulation: check with a health worker before sharing.

Corrected in verification: some calendars put the start on 6–7 February, so say "the Friday before Ramadan is expected". The Regional Chief Imam succeeded Sheikh Abdul Mumin Haroun, who died in December 2024. A 2020 COVID outreach through the Regional Chief Imam is reported but unverified (the page was blocked), so it is not relied on.

## 4. Keep the water pumps powered, and publish the cuts

PARTIALLY CONFIRMED | EMERGENCY MANAGER · HISTORIAN · SKEPTIC | ADAPTATION INVENTORY: NO PRECEDENT CONFIRMED

| OWNER | BY WHEN | COST | PREVENTS | PAYER / TRIGGER |
|---|---|---|---|---|
| **Ashanti Regional Minister, with ECG and Ghana Water Ltd (GWL)** | **priority feeders before 1 February 2027; schedule published before rationing starts** | **none: an allocation decision** | **the chain from dry taps to stored water to cholera, at the heat peak** | **ECG and GWL; trigger: the date** |

Give the Barekese and Owabi pumping stations priority feeders before February, and have GWL publish a rationing schedule. Barekese supplies over 70% of Kumasi's water. In February 2026, power outages cut the plants' output by up to 84% of operational capacity (the GWL managing director, reported by MyJoyOnline), and parts of the city went three days without water; production fell by about 959,000 m³ in March. GWL and ECG plan a dedicated transformer and an 11 kV substation upgrade. El Niño years have been low-Volta blackout years here: 1983–84, 1998, 2007, 2015, and 2024.

* Suggested formulation: check with a health worker before sharing.

Corrected in verification: the 84% figure is not in the DLF report, which describes ECG cable faults. A claim that Barekese is 40% silted has no source; the GWL managing director calls it "increasingly shallow". PURC ordered ECG to publish a load-shedding timetable in March 2024, and ECG answered that there would be none. The February 2026 cuts were faults and low voltage, not rationing, and no 2026 Kumasi timetable was found.

## 5. Stock the clinics for diarrhoea and cholera before the taps run dry

PARTIALLY CONFIRMED | HISTORIAN · PHILANTHROPIST · EMERGENCY MANAGER · ECONOMIST | ADAPTATION INVENTORY: NO PRECEDENT CONFIRMED

| OWNER | BY WHEN | COST | PREVENTS | PAYER / TRIGGER |
|---|---|---|---|---|
| Municipal Health Directorate, with the Aboabo and Asawase health facilities and Manhyia District Hospital | stocked by 31 January 2027 | GHS 195,000 (US$16,738) for ORS, zinc, and chlorine (placement 2); clinic heat surge GHS 80,000 (US$6,867), unverified (placement 4) | diarrhoea and cholera deaths when water is rationed and after the first rains | donor grant through the Directorate; release trigger: GWL rationing or a confirmed cholera case |

Pre-position ORS with zinc, chlorine tablets, and IV fluids, and run food-vendor hygiene, released when GWL rations supply or a cholera case is confirmed. ORS cuts diarrhoea deaths in children under five by about 69% (95% CI 51–80; Munos, Walker, and Black, 2010), and zinc by about 23% (Fischer Walker and Black, 2010). Ashanti's 2014 outbreak reached 203 cases and 2 deaths by November. In 2024–25, Ghana had 6,145 suspected cases and 49 deaths, but Ashanti had only 5 confirmed cases, all recovered. The IFRC cholera DREF MDRGH020, CHF 135,759 (about GHS 1.9 million, US$164,000), went to Central, Eastern, Greater Accra, Western, and Western North, not Ashanti.*

* Suggested formulation: check with a health worker before sharing.

Corrected in verification: the 69% figure pools three quasi-experimental studies in children under five; the 23% zinc figure is from Fischer Walker and Black, not Munos. Ashanti's 2014 count of 160 cases was a mid-September partial figure. No IFRC anticipatory or DREF instrument covers Ashanti or Kumasi; an imminent-crisis DREF must be requested by the Red Cross once the event is imminent and cannot be pre-agreed. No facility-level record for the Aboabo or Asawase clinics was found.

## 6. Desilt the Pelele and Asabi Junction drains by 15 February, haul the silt away in the same contract, and guard the crossings from the first March storm

PARTIALLY CONFIRMED · EMERGENCY MANAGER · HISTORIAN · ECONOMIST · PHILANTHROPIST · FRONTLINE · SKEPTIC

ADAPTATION INVENTORY: NO PRECEDENT CONFIRMED

| OWNER | BY WHEN | COST | PREVENTS | PAYER / TRIGGER |
|---|---|---|---|---|
| Asokore Mampong Municipal Assembly (the MCE, Ben Abdallah Alhassan) and its Works Department; the Municipal Security Council's police and fire task force and NADMO at the crossings | drains clear and silt removed by 15 February 2027; crossings guarded from the first March storm | desilting: no published figure, GHS 60,000–150,000 (US$5,200–12,900) is a lens estimate; flood wardens GHS 120,000 (US$10,300), placement 3 | drownings at the crossings in the first rains, the documented killer here | the Assembly (DACF and its own revenue), with a 1:1 matching grant for wardens; trigger: the calendar |

Clear the drains by 15 February, and write the haulage into the same contract, so the job is not done until the silt is at the disposal site. From the first March storm, station the police and fire task force (first deployed on 16 June 2026 for people "crossing fast-moving floodwaters") and NADMO at the known crossings with rope, lights, and barricades, and have youth volunteers mark the deep crossings in daylight. There is precedent: NADMO dredged in July 2017, the Zongo Development Fund dredged 1.7 km in April 2018, and the Assembly dredged the Pelele and Asabi Junction drains in May 2025. The MCE says there has been no flooding since; that is his word only. In June 2026 the Pelele drain still "needs full re-engineering", which is unfunded.

Corrected in verification: the 2026 DACF is GHS 8.77 billion (US$753 million) across 261 assemblies, a gross average of about GHS 33.6 million (US$2.9 million) before deductions, with 2024 arrears of GHS 7.33 billion (US$629 million), so releases are late and partial. No GHS 25 million minimum per assembly is published, so no cost is derived from it. The Assembly had collected GHS 543,374 (US$46,600)

* Suggested formulation: check with a health worker before sharing.

of its GHS 4.6 million (US$395,000) 2026 revenue target by the end of the first quarter. The forest-zone rains start in the second or third dekad of March; the 2008 deaths came on 10 March.

## 7. Reach the head porters where they sleep, at 6 pm, January–March

PARTIALLY CONFIRMED | FRONTLINE · EMERGENCY MANAGER | ADAPTATION INVENTORY: NO PRECEDENT CONFIRMED

| OWNER | BY WHEN | COST | PREVENTS | PAYER / TRIGGER |
|---|---|---|---|---|
| Ghana Red Cross Society (Ashanti) volunteers, with women volunteers who speak Dagbani or Hausa | sleeping places found by 15 January 2027; evening visits January–March | volunteer time, within placement 1 | heat illness among porters, and porters caught by a storm without warning | within the Red Cross grant; trigger: the date |

Bring water and ORS to the porters' sleeping places at 6 pm, when they gather to bathe, and collect one phone number for each group so it can receive flood and heat alerts. The first task is to find out where they sleep now: in 2015 about 150 kayayei slept in a wooden hostel at Aboabo for GHS 2 a night (about US$0.17 at today's rate; Graphic), and in January 2016 a fire destroyed the porters' shelter near Afful Nkwanta after the Assembly moved them from beside the drain. Ask the Sarkin Zongo. The 2008 drowning victims were two teenage porters walking home.

* Suggested formulation: check with a health worker before sharing.

Corrected in verification: the hostel figures are old (an undated Graphic feature), and the shelter's current status is unverified. The Kayayei Youth Association's Ashanti branch is documented for 2016 only.

## 8. Food: pay school-feeding caterers on time, and hold a mobile-money top-up ready on a price trigger

PARTIALLY CONFIRMED | FOOD · ECONOMIST | ADAPTATION INVENTORY: NO PRECEDENT CONFIRMED

| OWNER | BY WHEN | COST | PREVENTS | PAYER / TRIGGER |
|---|---|---|---|---|
| Ghana School Feeding Programme and the Ministry of Gender, Children and Social Protection (MoGCSP); the Department of Social Welfare with the Ghana Red Cross for mobile money; NAFCO and MoFA for a release rule | second-term grant paid by 31 January 2027; release rule written by 31 December 2026; top-up on the trigger | school feeding already budgeted; top-up GHS 1,000,000 (US$85,837), a government and Red Cross line released on the trigger | children losing their one guaranteed meal, and households losing cooked meals when prices rise | MoGCSP and the Red Cross; trigger: Ashanti inflation or Esoko Kumasi maize up 25% |

The school-feeding grant rose from GHS 1.50 to GHS 2.00 (US$0.17) per child per day; the 2026 budget is GHS 1.98 billion (US$170 million), but only 44% had been released by June. Pay the second-term grant by 31 January. Hold a mobile-money top-up of GHS 500 (US$43) for 2,000 households, GHS 1,000,000 (US$85,837), released when the trigger fires; in August 2026 the Red Cross paid GHS 1,200 (US$103) by mobile money to each of 1,050 flood-hit households. Ask NAFCO for a written Kumasi release rule. The food signal is weak and late: Ghana produced 4.6 Mt of maize in 2025 against 3.6 Mt of demand, white maize cost GHS 3,141 (US$270) a tonne in August 2026, 36% below a year earlier (AGRA), and national food inflation was 4.0% in September. But Ashanti's headline inflation was 9.8% against 5.2% nationally, so trigger on the regional figure.

Corrected in verification: the inherited 2024 caterer arrears were cleared, but local non-payment continued into 2026 (Krachi East caterers threatened a boycott in May 2026). NAFCO's funding is GHS 300 million (US$25.8 million): GHS 100 million in 2025 and GHS 200 million in 2026, the second not yet released by NAFCO's latest statement. It held 36,597 bags of maize at 31 December 2025 (about 20,400 t of all grains), before 50,000 bags were directed to ECOWAS on 26 September 2026. Its purpose is supply to institutions, released only "when necessary"; it has no published market-release rule. Southern Ghana's maize lean season and price peak run June–August (FAO GIEWS, WFP), not February–April, and GMet forecasts a late end to Kumasi's rains (early cessation only at Prang, Sunyani, and Akim

* Suggested formulation: check with a health worker before sharing.

Oda). Triggers: GSS monthly inflation (Ashanti line); Esoko's Kumasi Central Market prices, which reach the public irregularly through the media (no SRID Kumasi series is online); AGRA monthly maize, about six weeks late. LEAP pays GHS 380–630 (US$33–54) per two-month cycle from September 2026, but no precedent exists for paying a cycle early.

## 9. Get Kumasi into the next urban flood parametric tender, with Accra's fast pass-through written in

PARTIALLY CONFIRMED | ECONOMIST | ADAPTATION INVENTORY: NO PRECEDENT CONFIRMED

| OWNER | BY WHEN | COST | PREVENTS | PAYER / TRIGGER |
|---|---|---|---|---|
| Ministry of Finance, with NADMO and the Assembly; BMZ InsuResilience premium support | a long-term ask: the next tender; it cannot be placed before the March 2027 rains | not priced; Accra's premium support is GHS 23.3 million (US$2 million) over three years | flood money that arrives months after the flood | Ministry of Finance with BMZ; trigger: a satellite flood footprint over insured areas |

Ghana's satellite-triggered flood cover (IDF, UNDP-IRFF, and ISF, with ICEYE data) is designed for Greater Accra assemblies only. The insurer pays the Ministry of Finance within 10 days, and the Ministry pays NADMO within 3. BMZ and ISF premium support is GHS 23.3 million (US$2 million) over three years, covering 75%, then 50%, then 25%. Use the World Bank-supported Deltares study of the Aboabo, Wiwi, and Sisa basins and the INES vulnerability study of Aboabo and Asawase (commissioned August 2025) as the hazard layer. The pass-through matters: ARC's 2025 drought payouts, GHS 11.2 million (US$960,202) in February and GHS 22.5 million (US$1,927,334) in April, about GHS 33.6 million (US$2.89 million) in all, reached NADMO only at the end of September 2025, 5–7 months later (CSIS).

Corrected in verification: the Accra policies were released in late September 2025 and the premium is in the 2026 budget, but placement through a 2026 tender is only expected, so the cover is not in force. Kumasi is not covered. The Deltares and INES studies have published no outputs, and World Bank capital is only "future dialogue". The ARC payouts were for drought, not flood, and no 2026–27 ARC renewal was found.

* Suggested formulation: check with a health worker before sharing.

WHO DOES WHAT

## RACI CHART

**R** = Responsible (does the work) · **A** = Accountable (one per action; signs off and answers for it) · **C** = Consulted (asked before) · **I** = Informed (told after). Built from the owners named in the verified actions. No one here has been contacted or has agreed to a role; every row is a proposal.

| Action | Municipal Health Directorate | Manhyia, KATH, and local clinics | Ghana Red Cross (Ashanti) | Sarkin Zongo, imams, and Haske FM | Ashanti Regional Minister | ECG and Ghana Water | Asokore Mampong Municipal Assembly | NADMO and the police/fire task force | MoGCSP (School Feeding, Social Welfare) | NAFCO and MoFA | Ministry of Finance |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1. Named list and heat check | **A/R** | **C** | **R** | **R** | | | **C** | **I** | | | |
| 2. Daily death count | **A/R** | **R** | **I** | **R** | | | **I** | | | | |
| 3. Friday sermon on fasting in the heat | **C** | | **I** | **A/R** | | | | | | | |
| 4. Pumps powered, cuts published | **I** | | | | **A** | **R** | **I** | **I** | | | |
| 5. Clinic stock for diarrhoea and cholera | **A/R** | **R** | **I** | | | **I** | | | | | |
| 6. Drains desilted, crossings guarded | | | **I** | **C** | | | **A/R** | **R** | | | |
| 7. Head porters reached | **I** | | **A/R** | **C** | | | | **I** | | | |
| 8. School feeding and mobile-money top-up | | | **R** | | | | **I** | | **A/R** | **R** | **C** |
| 9. Kumasi in the flood parametric tender | | | | | **I** | | **C** | **R** | | | **A/R** |

What the chart shows: the Municipal Health Directorate is accountable for three of the nine actions, the health core, and the Red Cross does the work in three (actions 1, 7, and 8), including reaching the porters. Actions 2, 3, 4, and 7 cost little or nothing; they need decisions from the Directorate, the Regional Chief Imam, the Regional Minister, ECG, and Ghana Water, not money.

WHERE MONEY DOES THE MOST

## FUNDING

The Philanthropist lens, reconciled against the ranked actions and verified. Placements are ranked by marginal lives saved per dollar. The envelope is bounded by what this place can spend before the peak, not by what it needs. Household-based amounts assume about 7,800 households in Aboabo, which is unverified: no Aboabo-only population is published.

FLOOR

**GHS 270,000 (US$23,176)**

TARGET

**GHS 770,000 (US$66,094)**

CEILING

**GHS 1,445,000 (US$124,034)**

LARGEST UNCERTAINTY

**Aboabo's real household count, and the Red Cross branch's volunteer roster (no count found)**

* Suggested formulation: check with a health worker before sharing.

Floor = heat outreach (GHS 75,000) + ORS, zinc, and chlorine (GHS 195,000) = GHS 270,000 (US$23,176). Target = 75,000 + 195,000 + flood wardens and secondary drains (120,000) + clinic heat surge (80,000) + cash for 500 households (300,000) = GHS 770,000 (US$66,094). Ceiling = the target + 40 more volunteers (GHS 75,000) + 1,000 more cash households at GHS 600 (GHS 600,000) = GHS 1,445,000 (US$124,034). The target did not move in verification. Two sensitivities were not adopted: at Ghana's national average of about 3.6 people a household (from memory, unverified), Aboabo would have about 9,700 households and the ORS line would add GHS 47,500 (US$4,077), +6%; paying the Red Cross's current GHS 1,200 per household instead of GHS 600 would make the target GHS 1,070,000 (US$91,845), +39%. GHS 1,200 is the Red Cross's loss-recovery level; GHS 600 is kept because this is a heat top-up.

Outside the envelope: the GHS 1,000,000 (US$85,837) mobile-money food top-up in action 8 is a government and Red Cross line released on a price trigger, not philanthropy, and it uses the same channel and largely the same households as placement 5. The grain-bank money proposed by the food lens is dropped (see "Do not do this"). Already funded, and not duplicated: the Red Cross flood protocol MDRGH022, CHF 220,000 (about GHS 3.1 million, US$266,000), for the White Volta in the north; an EU allocation of EUR 80,000 (about GHS 1.04 million, US$89,000) for flood preparation in the north; and the Kumasi Resilient Project, at study stage with no capital. The Zongo Development Fund cannot be relied on: its CEO says it was left with GHS 221 (US$19), a partisan claim. No anticipatory heat money exists anywhere in Ashanti.

| RECIPIENT | USE | AMOUNT | MECHANISM | BASIS | VERDICT |
|---|---|---|---|---|---|
| Ghana Red Cross (Ashanti) with the Municipal Health Directorate | Actions 1 and 7: heat list and visits, 15 January – 31 March; 40 volunteers × 11 weeks × GHS 150 (US$12.88) = GHS 66,000, plus GHS 9,000 for airtime and thermometers | GHS 75,000 (US$6,438) | direct grant | Heat early-warning benefit-cost above 50:1 in Lucknow, Chennai, and Surat (World Bank Policy Research WP 10960; city-wide systems, not volunteer outreach). A figure of about GHS 291 (US$25) per DALY for Ahmedabad is from the EA Forum and is not peer-reviewed. GHS 150 a week is about 1.4× the statutory minimum wage | Partially confirmed |
| Municipal Health Directorate | Action 5: ORS with zinc, and chlorine, for about 7,800 households × GHS 25 (US$2.15), a unit-cost estimate; the household count is unverified | GHS 195,000 (US$16,738) | pre-positioned stock | ORS+zinc co-packs about GHS 2,190 (US$188) per DALY (Child Health Task Force 2024); ORS+zinc about GHS 30,000 (US$2,574) per death averted in Nigeria (GiveWell) | Partially confirmed |

* Suggested formulation: check with a health worker before sharing.

| RECIPIENT | USE | AMOUNT | MECHANISM | BASIS | VERDICT |
|---|---|---|---|---|---|
| Asokore Mampong Municipal Assembly | Action 6: flood wardens and secondary drains; 60 youth × 20 days × GHS 80 (US$6.87) = GHS 96,000, plus GHS 24,000 for tools and rope | GHS 120,000 (US$10,300) | 1:1 matching grant | No published cost-effectiveness; drowning at crossings is the documented harm. GHS 80 a day is 3.7× the GHS 21.77 (US$1.87) minimum wage; no Kumasi casual rate was found | Partially confirmed |
| Aboabo and Asawase health facilities | Action 5: clinic heat surge (fans, shade, IV fluids, cold water), 4 sites × GHS 20,000 (US$1,717) | GHS 80,000 (US$6,867) | direct grant | An absorption-led sum; no facility-level record was found | Unverified |
| Ghana Red Cross (mobile money) | Actions 1 and 8: cash for 500 high-risk households × GHS 600 (US$52), February | GHS 300,000 (US$25,751) | cash transfer | The Red Cross paid GHS 1,200 (US$103) per household by mobile money in August 2026, and GHS 600–1,200 in 2023; GHS 600 is a heat top-up only. No death-averted figure exists | Partially confirmed |
| Ghana Red Cross (Ashanti) | Actions 1 and 7: 40 more volunteers (ceiling only) | GHS 75,000 (US$6,438) | direct grant | As above; only if the branch shows it has the volunteers | Partially confirmed |
| Ghana Red Cross (mobile money) | Actions 1 and 8: 1,000 more cash households × GHS 600 (ceiling only) | GHS 600,000 (US$51,502) | cash transfer | As above; cross-check the list against LEAP households | Partially confirmed |

A funder's view: the gap here is not money but a named recipient in Aboabo who holds a signed trigger. GHS 75,000 (US$6,438) of volunteers arrives before the funerals; GHS 10 million (US$858,000) of concrete arrives after them. Leave the Pelele re-engineering to the World Bank and the Assembly.

* Suggested formulation: check with a health worker before sharing.

# Do not do this

CONTRAINDICATED · ADVICE THAT WOULD COST LIVES HERE

Advice that circulates for this hazard and would cost lives here.

## Stockpiling maize for February–April, in a community grain bank or at home

CONTRAINDICATED

Found in verification; it began as the food lens's first action. Southern Ghana's maize price peak runs June–August, not February–April, and prices are 36% below last year. NAFCO already holds stock in Kumasi and buys at a floor of GHS 450 (US$39) per 100 kg. A grain bank released in February would sell into a glut and undercut zongo traders, and it carries aflatoxin risk (grain above about 13% moisture) and a selection process open to capture.

**Instead:** the mobile-money top-up on a price trigger (action 8). A household that cooks TZ or banku at home may buy one bag in June–July, before the peak.

## Clearing the riverbank before the rains with no resettlement

DO NOT CONFIRMED

On 11 June 2015 the Assembly demolished more than 80 wooden structures at Pelele, 16 days before the 27–28 June flood, and no resettlement is mentioned. On 23 September 2025 it told riverbank residents along the storm drain to leave "willingly or [be] forced", with no compensation mentioned; whether demolitions followed is unverified.

* Suggested formulation: check with a health worker before sharing.

**Instead:** a written relocation and compensation offer agreed with the Sarkin Zongo, targeting only the structures a survey shows constrict the channel.

## Counting desilting as done while the silt sits on the bank

DO NOT PARTIALLY CONFIRMED

Eight months after the Odaw in Accra was dredged, the silt was still heaped on its banks and washing back in with each rain, and President Mahama warned against leaving silt on roadsides after the 2025 clean-up. No Kumasi case was found; this is Accra evidence.

**Instead:** haul the silt in the same contract, and photograph the empty bank (action 6).

## Copying the "no fans above 35 °C" rule into Kumasi's humid heat *

DO NOT CONFIRMED

At 38 °C and 60% humidity, fans and skin wetting lowered core temperature and cardiac strain in older adults, including those with coronary artery disease; fans did harm only at 45 °C and 15% humidity (Blanchet, Gagnon, and Jay, *NEJM* 2024). Morris and colleagues (2021) put the fan-safe limit below 38 °C for healthy older adults. Kumasi's February maximum is about 33–34 °C. This reverses the dry-heat advice in the Sahel briefings.

**Instead:** fan plus wet skin; a damp cloth when the power is off.

## Planning on trigger money arriving before the peak

DO NOT CONFIRMED

ARC's 2025 drought payouts took 5–7 months to reach NADMO. The Accra flood cover is not in force and excludes Kumasi, and no IFRC anticipatory instrument covers Ashanti.

* Suggested formulation: check with a health worker before sharing.

**Instead:** money already held by the Red Cross and the Assembly (see the funding section).

### Relying on cooling centres

DO NOT PARTIALLY CONFIRMED

Cooling centres are under-used even in wealthy cities, because people do not know about them, cannot reach them, or worry about safety and stigma. Here the power fails with the heat: in January 2024 ECG blamed the heat for outages.

**Instead:** action 1's home visits and shaded yards.

### Storing water without a lid

DO NOT CONFIRMED

Uncovered stored water breeds *Aedes* mosquitoes and carries a cholera risk, and the pump cuts will push households to store water.

**Instead:** covered, scrubbed containers (For you, step 1).

### Adding ceiling insulation for relief on hot nights

DO NOT PARTIALLY CONFIRMED

In 47 rooms in Accra and Tamale, insulated ceilings made rooms 1.4 °C warmer at night, though 6.9 °C cooler by day (Wilby and colleagues, 2021). It is a trade-off, not pure harm, but it is the wrong fix for hot nights.

* Suggested formulation: check with a health worker before sharing.

**Instead:** for night relief, ventilate; check how hot the room is at night.*

### Buying heat-income insurance

DO NOT CONFIRMED

No heat-income cover is sold to urban workers in Ghana; the one retail parametric product is QIC's farm cover with Ibisa, launched in December 2024. The precedent is Indian: SEWA's scheme pays GHS 23–233 (US$2–20) a member, a day or two of wages.

**Instead:** fund the heat list and visits (action 1).

### Crossing moving water, or running across a drain in a storm

DO NOT CONFIRMED

On 10 March 2008 two teenage porters drowned in an Aboabo drain, the second trying to save the first. The 2026 task force was deployed because people were "crossing fast-moving floodwaters".

**Instead:** pick the crossing in daylight, and wait out the storm (For you, step 4).*

WHAT HAS HAPPENED HERE BEFORE — THE RECORD

| YEAR | EVENT | TOLL | WHAT CHANGED AFTER |
|---|---|---|---|
| 1982–84 (strong El Niño 1982–83) | Drought, bushfires, and famine; Akosombo power rationing, with Volta inflow below 15% of normal | "Hundreds" nationally (Wikipedia) | Not recorded |
| January 1998 (strong El Niño 1997–98) | Power supply cut by 30% as the Volta fell; Akosombo output down from 900 to 360 MW | n/a | Not recorded |
| October 1999 (La Niña) | Cholera in Kumasi; public food sales banned | Not verified (page blocked) | Food-sales ban, unverified |

* Suggested formulation: check with a health worker before sharing.

| YEAR | EVENT | TOLL | WHAT CHANGED AFTER |
|---|---|---|---|
| 2007 (weak El Niño 2006–07) | Volta lake at an all-time low; load-shedding | n/a | Not recorded |
| 10 March 2008 (La Niña) | First heavy rain: two teenage kayayei fall into a deep Aboabo drain while running for shelter | 2 (one died in hospital; one body never found) | NADMO said it was "not aware" |
| September–November 2014 (neutral) | Cholera in Ashanti, during Ghana's worst outbreak since 1982 | 203 cases, 2 deaths | Not recorded |
| February 2015 (developing El Niño) | Akosombo at 244 ft; a 450 MW deficit (dumsor) | n/a | Not recorded |
| 11 June 2015 (El Niño) | The Assembly demolishes 80+ wooden structures at Pelele, on the stream | None reported | No resettlement mentioned |
| 27–28 June 2015 (El Niño) | Aboabo river floods during dumsor; residents trapped for more than 12 hours | At least 1 (Graphic); local radio (Citi) reported 5 | Youths burned tyres; President Mahama promised desilting and a new bridge "soon" (1 July 2015) |
| July 2017 (neutral) | NADMO dredges the Aboabo drains | n/a | Residents warned it was futile without new bridges |
| April 2018 (neutral) | Zongo Development Fund dredges 1.7 km of the Aboabo storm drain | n/a | Not recorded |
| 24 June 2021 (neutral, after La Niña) | Kumasi floods, with deaths at Aboabo and Asabi Junction; a toddler swept away at Sawaba | 7 across Kumasi (NADMO) | Not recorded |

* Suggested formulation: check with a health worker before sharing.

| YEAR | EVENT | TOLL | WHAT CHANGED AFTER |
|---|---|---|---|
| March 2024 (decaying El Niño) | Outages across Ghana; PURC orders ECG to publish a load-shedding timetable | n/a | ECG said there would be no timetable |
| December 2024 (weak La Niña) | Cholera in Ashanti during the national outbreak | 5 confirmed cases (1 in Kumasi), 0 deaths | Regional emergency operations centre and case searches |
| May 2025 (neutral) | The Assembly dredges the Pelele and Asabi Junction drains | n/a | The MCE says no flooding since (his word only) |
| 23 September 2025 | The Assembly tells riverbank residents to leave "willingly or [be] forced" | n/a | No compensation mentioned; demolitions unverified |
| February–March 2026 | Power outages cut Barekese and Owabi output by up to 84% (GWL); parts of Kumasi without water for three days | n/a | GWL and ECG plan a dedicated transformer and an 11 kV upgrade |
| 16 June 2026 (developing El Niño) | The Assembly deploys a police and fire task force at flood crossings | n/a | The Pelele drain still "needs full re-engineering", unfunded |

One item reported elsewhere was removed: a "July 2024" flood with seven drowned at Asabi Junction and Aboabo is the June 2021 flood re-dated, because the Pulse story repeats the 2021 wording and its "2024…" URL stamp is an archive-migration timestamp, not the event date.

The recurrence nobody has noticed: El Niño years are blackout years here (1983–84, 1998, 2007, 2015, and 2024), and the drownings come in the dark at the first rains. Every recorded death in Aboabo is a drowning, because Kumasi has never counted a heat death; the forecast 1,000 this winter would go uncounted too. ENSO phases in this table are our assessment against NOAA's ONI record, not sourced per event.

WHERE THE EXPERTS DISAGREE — UNRESOLVED

- **Ashanti faces 1,100 heat deaths** (the Climate Impact Lab forecast) versus **Ghana's own data show risk at both extremes, clearest for women in the cold** (Public health, from Kintampo: cold RR 1.61 at 18 °C; heat RR 1.71 at 31 °C, 2.19 in males, null in females; women's excess is cold-driven, RR 4.88 at 18 °C). Action 2's daily count is what would settle it.

* Suggested formulation: check with a health worker before sharing.

- **The flood agenda, with its bulldozers, takes the attention while heat kills quietly** (Skeptic) versus **the documented deaths are drownings at the first rains** (Emergency manager, Historian). Both are kept: heat is ranked first, because it peaks first, and the drains sixth, because they matter from March.
- **Buy forward into a community grain bank at glut prices** (Food) versus **the price peak is in June–August and NAFCO already holds stock** (food verifier). The grain bank is contraindicated ("Do not" 1).
- **Do not use fans in the heat** (the dry-heat rule used in the Sahel briefings) versus **fans plus wet skin help in humid heat** (*NEJM* 2024). Kumasi's humid heat reverses the rule.

## WHAT NOBODY COVERED — BLIND SPOT

No lens covered three things. First, the night-time temperature inside Aboabo's compound rooms: no measurement exists, and it decides whether a fan and a damp cloth are enough. Second, people with sickle-cell disease: no Kumasi study links heat to their admissions, though about 1 in 8 newborns carried the trait. Third, whether Aboabo's head porters still sleep in the neighbourhood at all, which decides whether action 7 reaches anyone.

## HOW TO READ THIS — METHOD

**The panel is author-built.** Eight simulated expert lenses were used to generate these recommendations. Where they agree, that is a strong hypothesis, not a consensus of the field. Every recommendation was then independently checked for evidence, transferability to this location, and feasibility before the peak; the verdict tags carry the result.

Mortality figures are published at region scale, for the Ashanti Region. They are not figures for Aboabo or the municipality. The El Niño projection covers *heat only*; drought, flood, fire, smoke, and crop failure are additional to it and are carried here by the historical record rather than by a published number.

## HOW THIS BRIEFING WAS PRODUCED — METHODOLOGY

- **Scope.** The location was taken as given: Aboabo, a zongo neighbourhood of Kumasi along the Aboabo stream, at about 6.699°N, 1.598°W (OpenStreetMap). It is made up of the Aboabo No. 1 and No. 2 electoral areas, two of the ten electoral areas of Asokore Mampong Municipal

* Suggested formulation: check with a health worker before sharing.

Assembly, which was carved from the Kumasi Metropolitan Assembly in 2012; the MCE is Ben Abdallah Alhassan, confirmed in May 2025. The municipality had 191,402 people at the 2021 census (via citypopulation.de); no Aboabo-only population is published, so household-based amounts assume about 7,800 households, which is unverified. The actions are built for the municipality and Kumasi. Mortality figures exist only at region level, for the Ashanti Region.

- **Exposure.** Climate Impact Lab El Niño Heat Mortality Tracker data, file `v261026_adm1_output_BASE_rounded_cust.csv` (version of 26 October 2026, pre-peer-review): Ashanti Region (GADM), 71 additional heat deaths June–September 2026, 1,000 forecast October 2026 – March 2027 (October 40, November 86, December 47, January 160, February 360, March 340), 1,100 in total, the highest of Ghana's 16 regions (Northern is second, at 1,000); Ghana, 7,600 (7,100 October–March); the world, 857,400. This update supersedes the September report *451,000 at Risk*. Method: ECMWF SEAS5 seasonal forecasts applied to temperature–mortality relationships against a 1996–2025 baseline. The figures count heat only. They are never added to the Impact Map's long-run projections. The figure is model-driven: Kintampo's surveillance data (2005–15, 10,865 deaths) show death risk rising at both cold and hot extremes.
- **Expert panel.** Eight simulated expert lenses (emergency manager, public health scientist, frontline responder, skeptic, economist, historian, philanthropist, and food supply chain analyst) each researched this place independently and in parallel, from web sources, to a fixed brief.
- **Reconciliation.** Duplicate actions were merged and backing lenses recorded. Convergence was tested for shared bias (whether the evidence behind an agreed action came from a setting like this one) before it counted in an action's favour. Contradictions were kept, not resolved by dropping a side. Actions were ranked by expected lives saved per unit of effort before the peak, each was given a named payer and trigger or marked unfunded, and the funding placements were reconciled against the ranked list.
- **Verification.** Five verifier agents (financial instruments; health; community, flood, power, and water; food; the funding envelope) checked every recommendation on three tests: evidence (the primary source and its effect size), transferability (whether that evidence came from somewhere like here), and feasibility (whether it can be done here before the peak). Financial actions were also checked for availability in this jurisdiction, a named payer, a trigger threshold, and whether they already exist; food actions for published triggers and timing. The funding envelope was checked for real recipients, published unit costs, and duplication, then re-summed.
- **Who does what.** The RACI chart is built from the owners named in the verified actions, with one accountable party per action. It is a proposal: no organisation named in it has been contacted or has agreed to a role.
- **Verdicts.** *Confirmed*: evidence, transferability, and feasibility all hold. *Partially confirmed*: holds after the corrections shown. *Unverified*: could not be checked to a primary source. *Contraindicated*: would do harm or waste the window here.
- **Adaptation Inventory.** Not usable in this run. The inventory page loaded, but it renders in the browser, its programme counter read 0, and it has no keyword search, so it returned no entries for Ghana and no precedent could be confirmed there. No action here should be read as

* Suggested formulation: check with a health worker before sharing.

having no precedent; none was checked.

- **Money.** Costs are shown in Ghana cedis (GHS) with US dollars in parentheses, at GHS 11.65 = US$1, the Bank of Ghana interbank mid-rate published 26 October 2026 (buying 11.6442, selling 11.6558), as reported by ModernGhana and WikiFX. Forex bureaus charge more, about GHS 12.25. Figures published in Swiss francs (the Red Cross protocols) are converted at about CHF 1 = US$1.21 (late October 2026, approximate), and the one euro figure at the ECB reference rate of €1 = US$1.1172 (30 October 2026).
- **Reader.** No role was given, so the For you section is written for a resident. Its steps were taken from the verified actions and from what the panel found about households here. Any step not drawn from a ranked action was verified on the same three tests. The rest of the report is written for the Assembly and its partners.
- **Language.** This briefing is in English only. English is Ghana's official language and the working language of the Assembly, NADMO, and the Ghana Health Service. Aboabo's home languages are split between Asante Twi and Hausa, with Dagbani, Mooré, and others, so no single translation would serve the neighbourhood. Action 3 puts the spoken messages into Hausa and Twi instead.
- **Health wording.** Passages where a wording error could hurt someone (ORS and zinc, water treatment, cooling and fans, fasting and the medical exemption, heat in pregnancy, danger signs, flood crossings, and the 112 emergency number) are marked "* Suggested formulation: check with a health worker before sharing." They were checked against published sources, not by a health professional. They are suggestions, not medical advice, and anyone passing them on should confirm them with a doctor, pharmacist, nurse, or community health worker first. The marker was added on 2 November 2026.
- **Rules.** No cost is derived by dividing a published total, and no local figure is invented; where data is thin, the report says so and shows the containing geography's number with its arithmetic. Every source is listed below with its verification status.
- **Limits.** The panel and verifiers are AI research agents working from web-available sources, not human experts, and no one in this place was interviewed. No Aboabo population is published, and the Red Cross Ashanti branch's volunteer roster was not found. The B&FT article on Barekese would not load. The Pulse "2024" flood item turned out to be a re-dated 2021 report. GhanaWeb blocked one fetch (the 2020 zongo COVID outreach), and the New Humanitarian 1999 cholera page could not be opened. No SRID Kumasi price series is online. The CHF rate is approximate. The locator map shows the neighbourhood at country scale. This briefing was run in parallel with one for Dioulassoba, Bobo-Dioulasso (Burkina Faso) in the same session.

References · status after verification

Climate Impact Lab, El Niño Heat Mortality Tracker data v261026 (downloaded 30 Oct 2026); tracker · confirmed

Place and governance: citypopulation.de, Asokore Mampong; ghanadistricts, electoral areas (page returned 500); Ghana Remembers, Aboabo No. 1 and

* Suggested formulation: check with a health worker before sharing.

No. 2; ghanadistricts, information van; Citi, MCE confirmed; GNA, MCE, Dec 2025; OpenStreetMap Nominatim, coordinates; GNA, Assembly revenue · confirmed (Aboabo population and household count: unverified)

Heat and health: Kintampo HDSS (PMC7527893); Navrongo (PMC3508691; heat-dominant in the north, intervals inconsistent); Hess et al. 2018, Ahmedabad; Tannor et al., Ashanti hypertension; PMC8801650, Kumasi screening; Cappuccio, central Ghana villages; GhanaWeb, Kumasi newborn screening; PMC8908904, KATH sickle-cell registry; MyJoyOnline, GMet heat advisory 2024; Graphic, Ghana Health Service heat advice; MyJoyOnline, chief forecaster on heatwaves; MyJoyOnline column, heat and dumsor (anecdote, unverified); Blanchet, Gagnon, and Jay, NEJM 2024; News-Medical on the NEJM letter; Morris et al. 2021; Weather Spark, Kumasi climate; weather2travel, Kumasi; Wilby et al. 2021, Accra and Tamale rooms; Nouna cool-roof trial protocol (no results); PMC9378433, cooling-centre use; Hanson et al., Nature Medicine 2024; The Standard on Hanson et al.; GNA, free maternal care; Kayayei health needs; medRxiv, Mogadishu burial surveillance; Jakarta Post, counting burials in Africa · partially confirmed (Ghana burial-count precedent: none found; age shares: unverified)

Diarrhoea and cholera: Munos, Walker, and Black 2010; Graphic, 2024–25 cholera; WASH Ghana, 2014 cholera; GhanaWeb, 2014 toll; ModernGhana, Ashanti cases Dec 2024; ClassFM, Ashanti response; ModernGhana, Kumasi cholera c.2005 (year unconfirmed); The New Humanitarian, 1999 (not opened); IFRC MDRGH020; PMC5381031, Aedes in stored water · confirmed (1999 outbreak: unverified)

Ramadan and community channels: Asaase Radio, Regional Chief Imam; GNA, ZOLAOD 2022; GNA, Tema heat advisory 2026; NCA, FM licences (Haske FM); Haske FM 91.7 listing; GhanaWeb, 2020 zongo COVID campaign (blocked); GhanaWeb, Zongo Fund and imams 2021; Graphic, Kumasi's head porters; Graphic, porters' shelter fire 2016; Graphic, NADMO 112 · partially confirmed (2020 outreach and porters' current sleeping places: unverified)

Power and water: MyJoyOnline, Barekese and Owabi (84%, GWL MD); MyJoyOnline, March 2026 production; DLF, Feb 2026 (no 84% figure); B&FT, siltation (body did not load; 40% unverified); Citi, no timetable 2024; GhanaWeb, PURC order; GhanaWeb, ECG blames heat 2024; MyJoyOnline, ECG Ashanti; Energy, Sustainability and Society, Volta inflows; GhanaWeb, 1998 cuts; The New Humanitarian, 2007 lake low; Peace FM, 2015 deficit; Graphic, PURC tariffs from 1 July 2026; MyJoyOnline, Q4 tariffs; GNA, Q2 tariffs (superseded) · partially confirmed (2026 Kumasi timetable: none found; Barekese siltation share: unverified)

Floods, drains, and demolitions: MyJoyOnline, 10 Mar 2008; Graphic, Aboabo flood 2015; Graphic, 28 Jun 2015 ("lights were off": unverified); Graphic, Pelele demolitions 2015; Citi, Aboabo-Asawase demolitions; Graphic, Mahama promise 2015; GhanaWeb, NADMO dredging 2017; MyJoyOnline, ZDF dredging 2018; Graphic, Kumasi floods 2021; Dubawa, 24 Jun 2021; ModernGhana, 2021 flood; Citi, 2021 toll; News Ghana, Sawaba toddler; Pulse "2024" item (contraindicated: the 2021 flood re-dated; removed); MyJoyOnline, Pelele 2022; MyJoyOnline, task force and dredging 2026; GNA, Sep 2025 notice; Daily Guide, ultimatum (one-week deadline unverified); GNA, 2026 market demolitions; 3News, Old Fadama 2015; Sage, forced removal in Accra; MyJoyOnline, Odaw silt; ModernGhana, Mahama on silt; GhanaWeb, Accra floods 1997; MDPI Climate, forest-zone rain onset · partially confirmed (2025 demolitions carried out: unverified; MCE's "no flooding since": his word)

Food: MyJoyOnline, AGRA prices Aug 2026; GNA, September inflation; MyJoyOnline, Esoko Kumasi prices; GMet, SON 2026 forecast; FAO GIEWS, Ghana; GNA, 50,000 bags to ECOWAS; MyJoyOnline, NAFCO stock; Graphic, NAFCO 2026 budget; News Ghana, NAFCO awaits GHS 200M; GhanaWeb, floor prices; MyJoyOnline, 2024 lean-season sack price; Adom, 2024 maize; Milling MEA, 2024 drought; MyJoyOnline, Ejura prices; DOAJ, hermetic-bag economics; ghanadistricts, Kumasi food sources; ISD, rice self-sufficiency; MyJoyOnline, El Niño and Ghana; Mensah et al. 2013, street food (garbled source); GNA, school feeding 2026; Graphic, caterer arrears cleared; GNA, Krachi East caterers; Adom, school-feeding budget · partially confirmed (grain bank: contraindicated; hermetic-bag price and SRID Kumasi series: not found)

Social protection and cash: GNA, LEAP rates from Sep 2026; MoGCSP, 96th LEAP cycle; GNA, 2025 LEAP increase (superseded); Citi, 99th LEAP cycle; GNA, SEND on the 2026 budget; GNA, LEAP mobile-money pilot; Graphic, Red Cross mobile-money cash 2026; MyJoyOnline, Red Cross flood response; MyJoyOnline, minimum wage · confirmed (early LEAP payment: no precedent found)

* Suggested formulation: check with a health worker before sharing.

Funding, finance, and insurance: CSIS, disaster risk financing in Ghana; Graphic, Kumasi flood studies; MyJoyOnline, housing minister on Kumasi floods; Ghanaian Times, Kumasi flood strategy; IFRC MDRGH022 sEAP; ReliefWeb, sEAP annual report; MyJoyOnline, EU EUR 80,000; GNA, Red Cross cholera volunteers (region-level only); Graphic, 2026 DACF; Citi, 2025 DACF (GHS 25M minimum: unverified); GhanaWeb, Zongo Development Fund balance; World Bank WP 10960; EA Forum, heat action plans (not peer-reviewed); Child Health Task Force, ORS+zinc co-packs; GiveWell, ORS+zinc in Nigeria; GiveWell, CHAI Bauchi; Charity Entrepreneurship (secondhand; superseded by GiveWell); Graphic, QIC weather-index insurance; GhanaWeb, insurance penetration; MyJoyOnline, remittances; GMet, Ashanti visit 2026; ValutaFX, CHF rate (August); ECB reference rate; Bank of Ghana interbank rate, 26 Oct 2026, as reported by ModernGhana and WikiFX · partially confirmed (Accra cover placement and 2026–27 ARC cover: unverified; clinic heat-surge sum: unverified)

History: Wikipedia, 1983 famine in Ghana · partially confirmed (ENSO phases: our assessment)

Exposure data · Climate Impact Lab, 451,000 at Risk · Impact Map · Adaptation Inventory

Generated 30 October 2026 for Aboabo, Kumasi. El Niño Ready is volunteer-run and is not affiliated with Climate Week NYC, The Climate Group, or Columbia University.

* Suggested formulation: check with a health worker before sharing.